

_FDSN data QC

Luděk Vecsey

Institute of Geophysics of the Czech Academy of Sciences, Prague

Station SoH

Using quality metrics & tools provided by archives:

EIDA

- EIDAWS-WFCCatalog, FDSNWS-Availability
- tools: Data Quality

<https://orfeus-eu.org/data/eida/quality/>

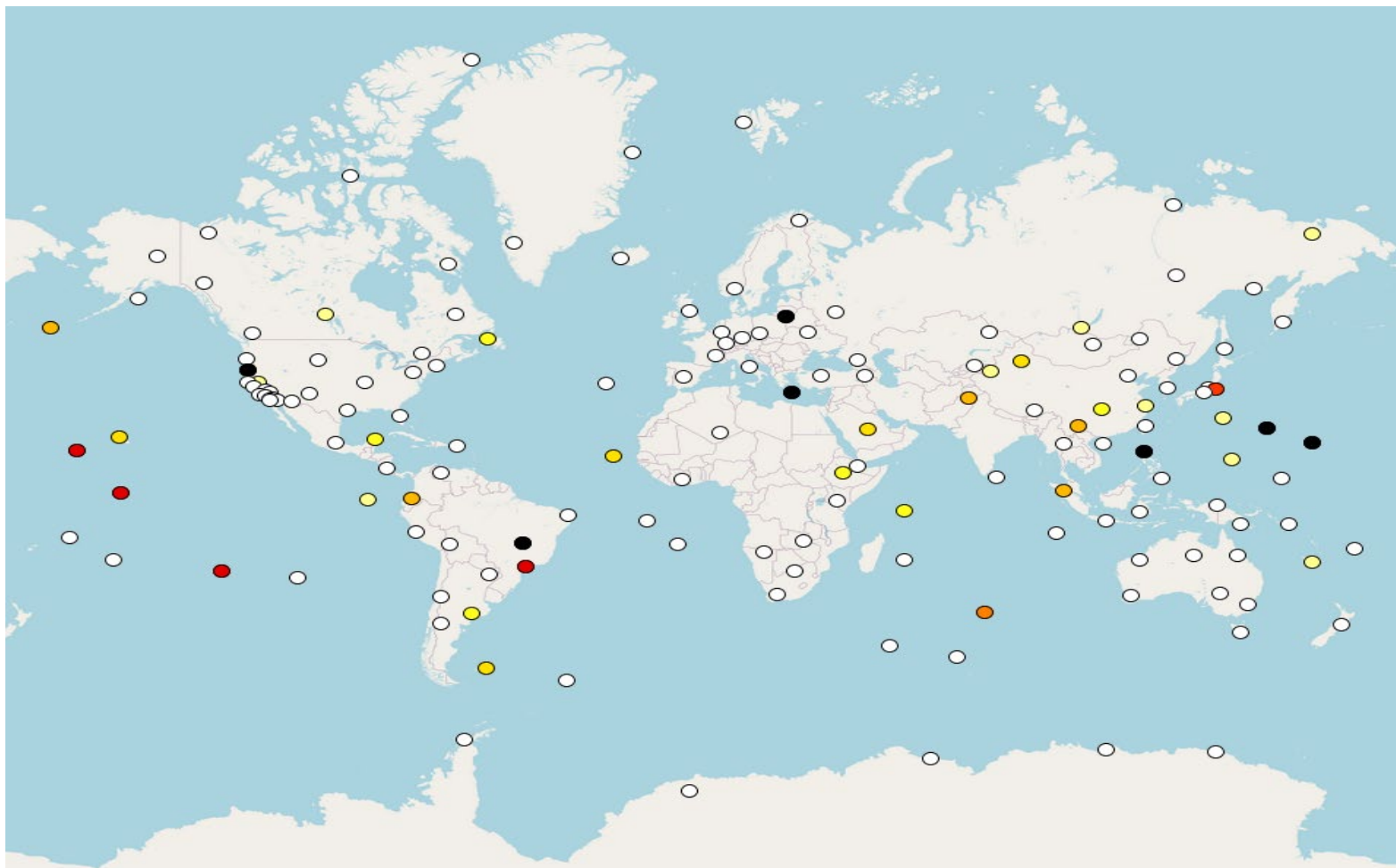
EarthScope

<https://service.earthscope.org/mustang/>

- MUSTANG (metrics, noise PSD/PDF)
- tools: Mustangular, MUSTANG Databrowser, Noise PDF Browser, ...

_FDSN data availability 2024+

[https://ds.iris.edu/mustang/mustangular/#/map?net= FDSN&cha=*HZ&start=2024-01-01&end=2025-08-29&metric=PERCENT_AVAILABILITY&coloring=hot&binmin=0&binmax=100&bincount=10&displayValue=Average&colocatedType=aggregate&aggregateValue=Maximum&channels=10.BHZ,00.BHZ,--.MHZ,--.LHZ,--.HHZ,--.BHZ,02.HHZ&absValue=false&invert=false](https://ds.iris.edu/mustang/mustangular/#/map?net=FDSN&cha=*HZ&start=2024-01-01&end=2025-08-29&metric=PERCENT_AVAILABILITY&coloring=hot&binmin=0&binmax=100&bincount=10&displayValue=Average&colocatedType=aggregate&aggregateValue=Maximum&channels=10.BHZ,00.BHZ,--.MHZ,--.LHZ,--.HHZ,--.BHZ,02.HHZ&absValue=false&invert=false)



Data availability		
Key		
On/off	Range	Count
	< 0.00	0
	0.00 - < 10.00	7
	10.00 - < 20.00	0
	20.00 - < 30.00	4
	30.00 - < 40.00	1
	40.00 - < 50.00	1
	50.00 - < 60.00	5
	60.00 - < 70.00	5
	70.00 - < 80.00	6
	80.00 - < 90.00	10
	90.00 - 100.00	117
	> 100.00	0

Daily means – fast and powerful tool for QC

Probabilistic power spectral density (PPSD)

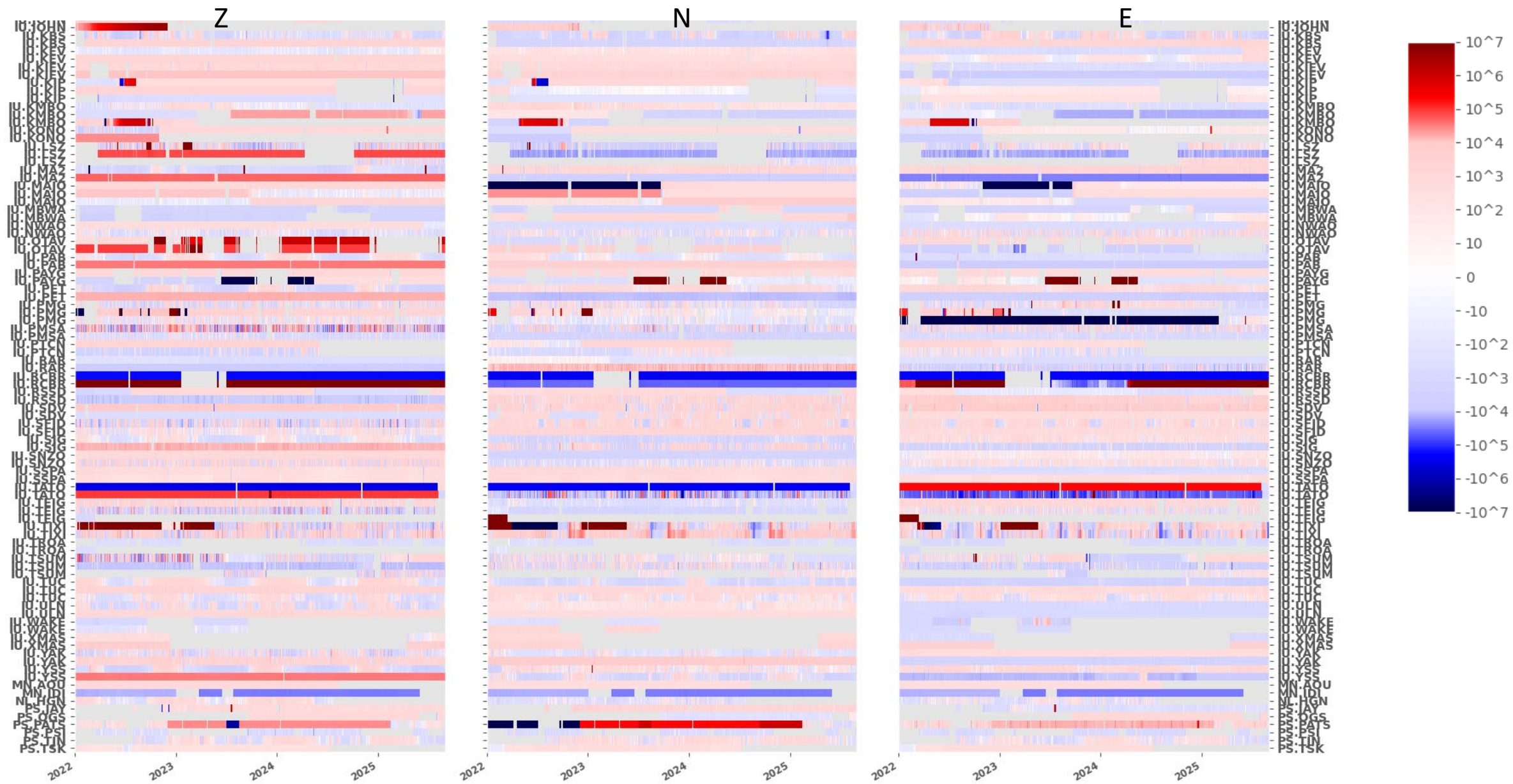
- time-frequency representation of the signal
- not included in EIDA WFCatalog (yet)
- need to download continuous data
- long computation of PPSD
- depends on metadata (changes/updates!)

Metrics in WFCatalog, Mustang

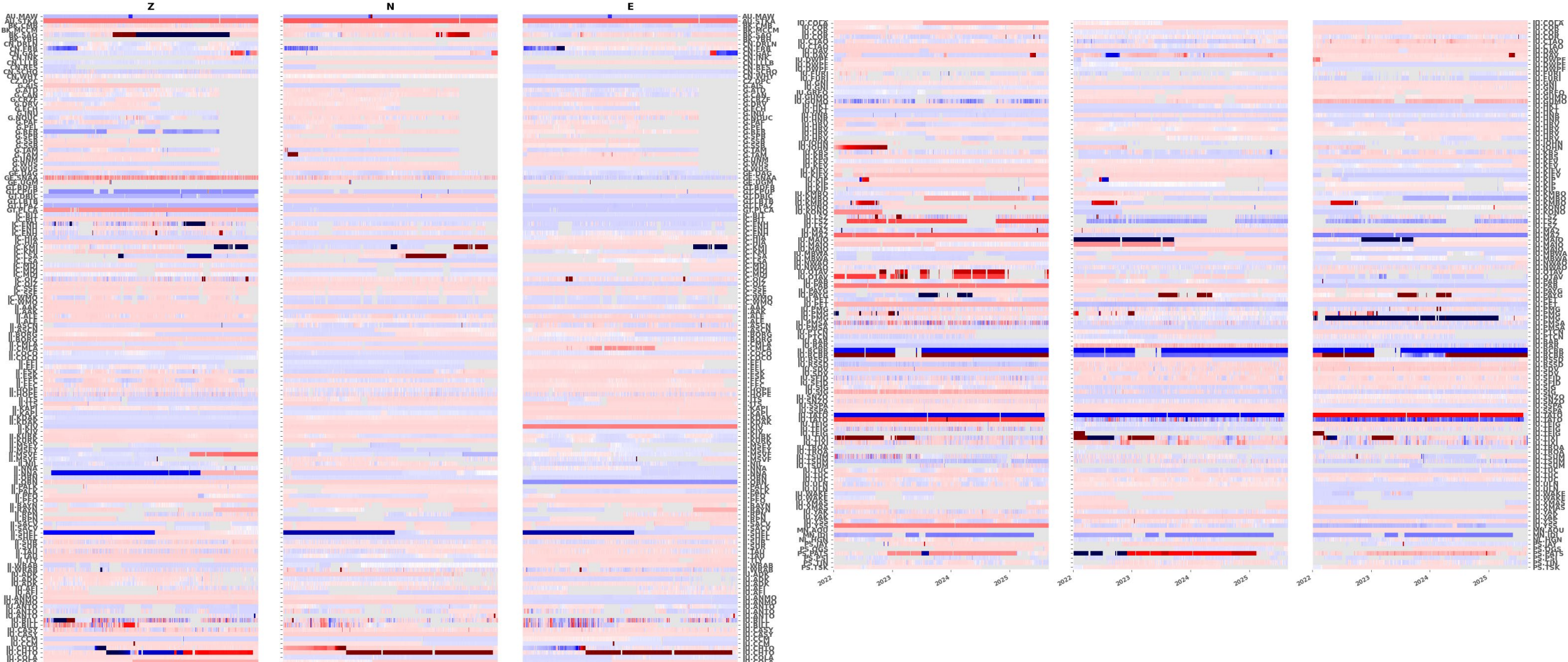
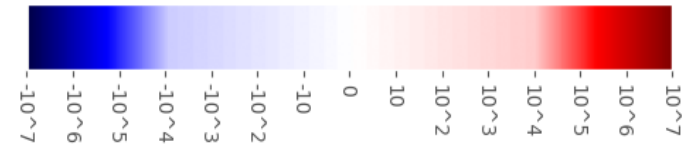
- no frequency analysis
- easy and fast to download from EIDA
- doesn't depend on metadata
- metrics:
 - daily median
 - daily std
 - daily max, min

...

Saturation of signal daily medians/offsets

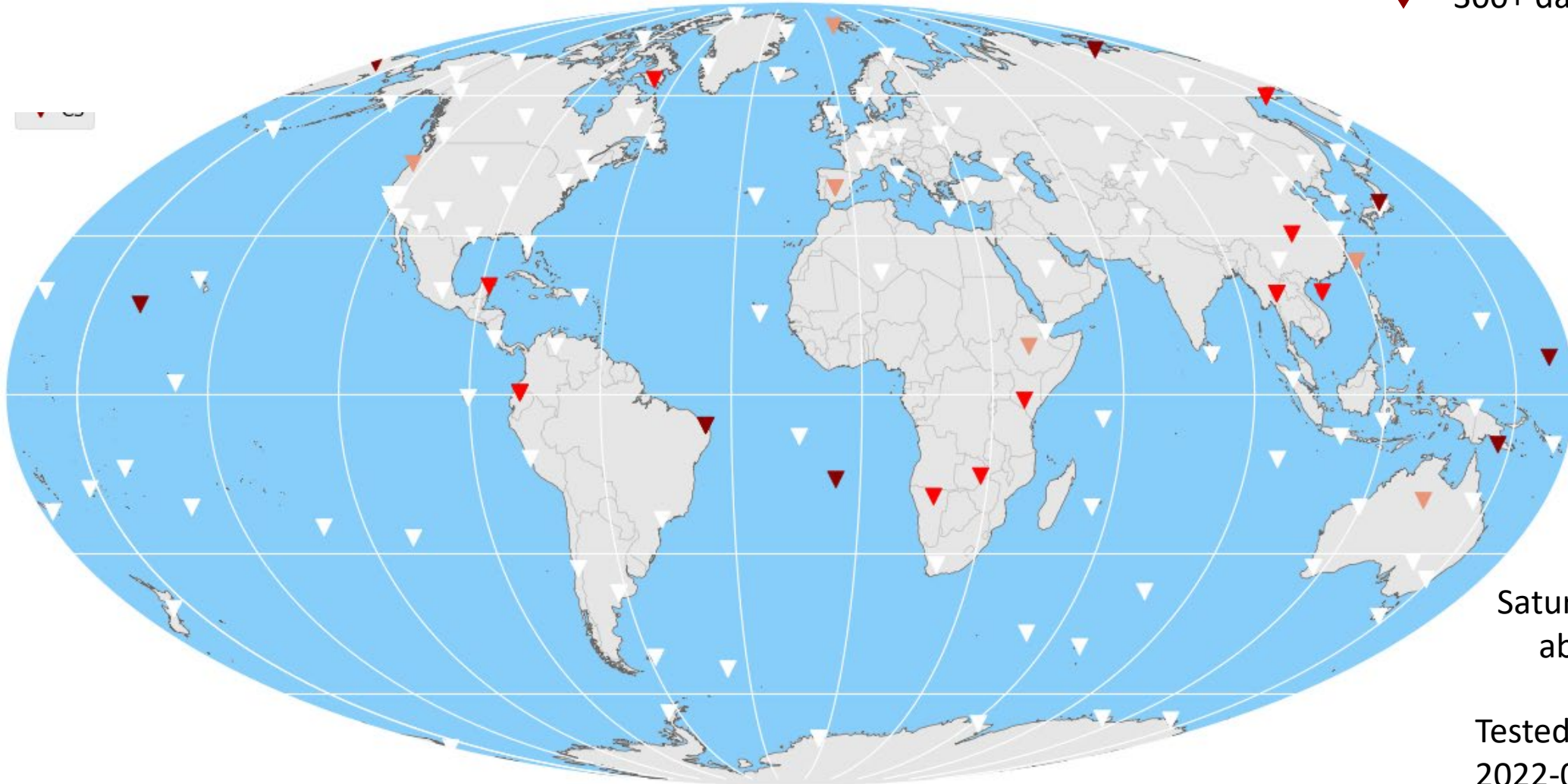


_FDSN - daily medians



Saturated medians - map

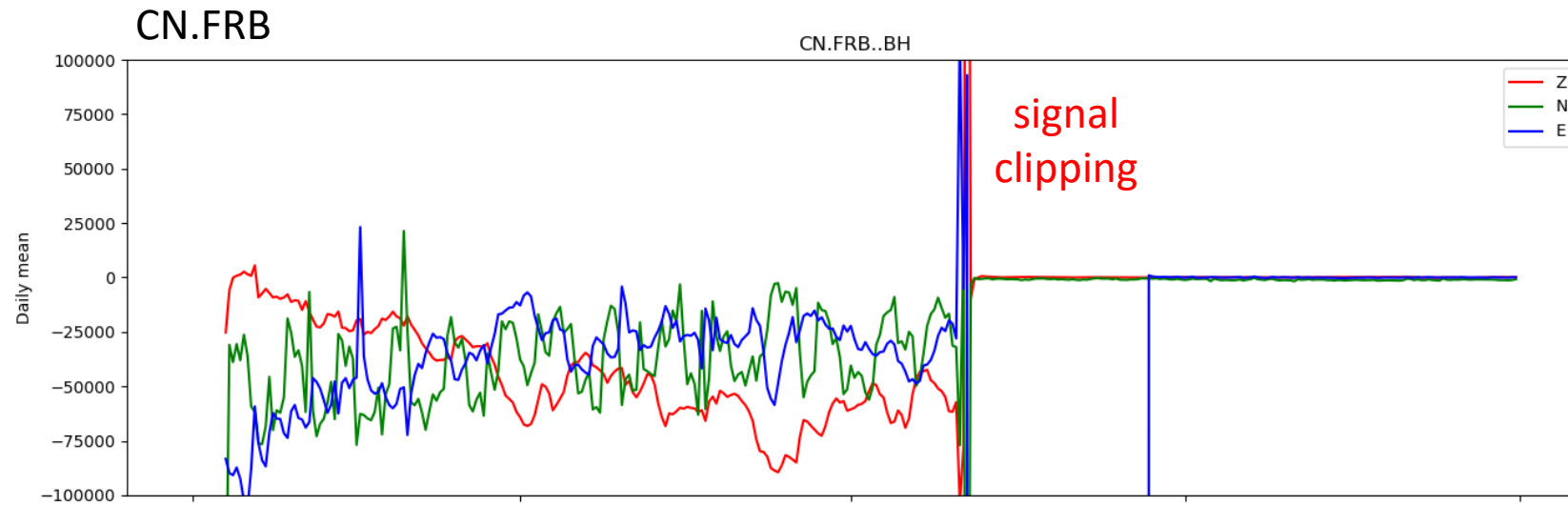
- ▼ all stations (137)
- ▼ 10+ days: 24/137 18 %
- ▼ 100+ days: 18/137 13 %
- ▼ 300+ days: 8/137 6 %



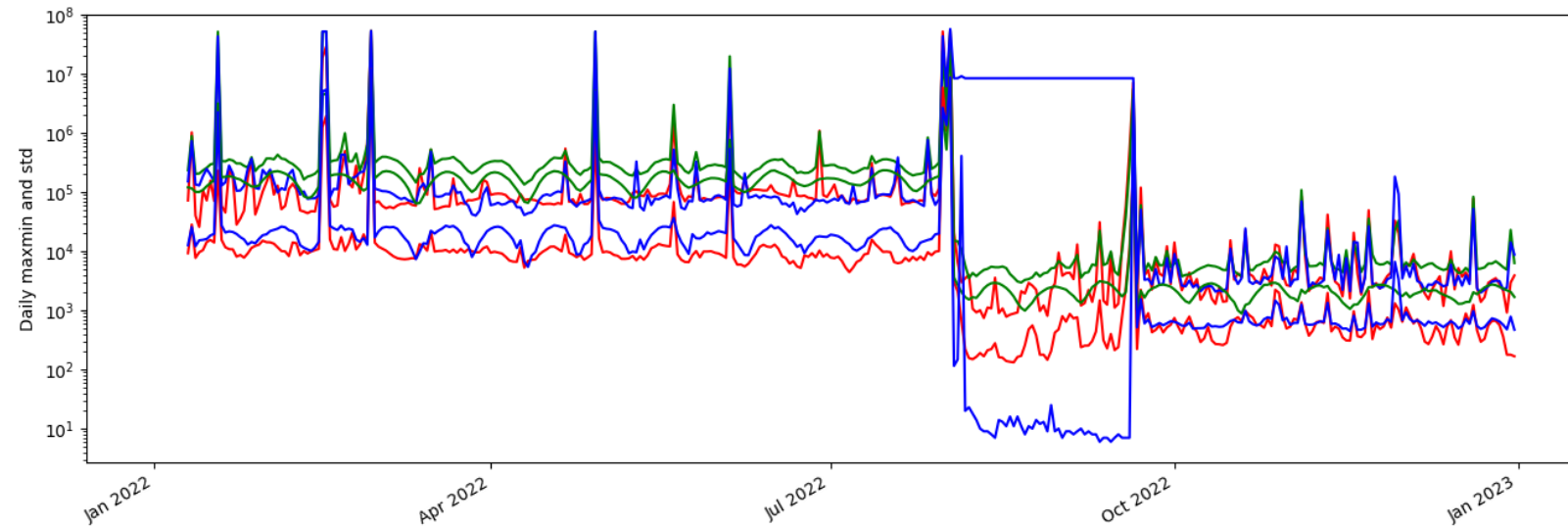
Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

Tested period:
2022-01-01 – 2023-01-01

Inspection of individual station

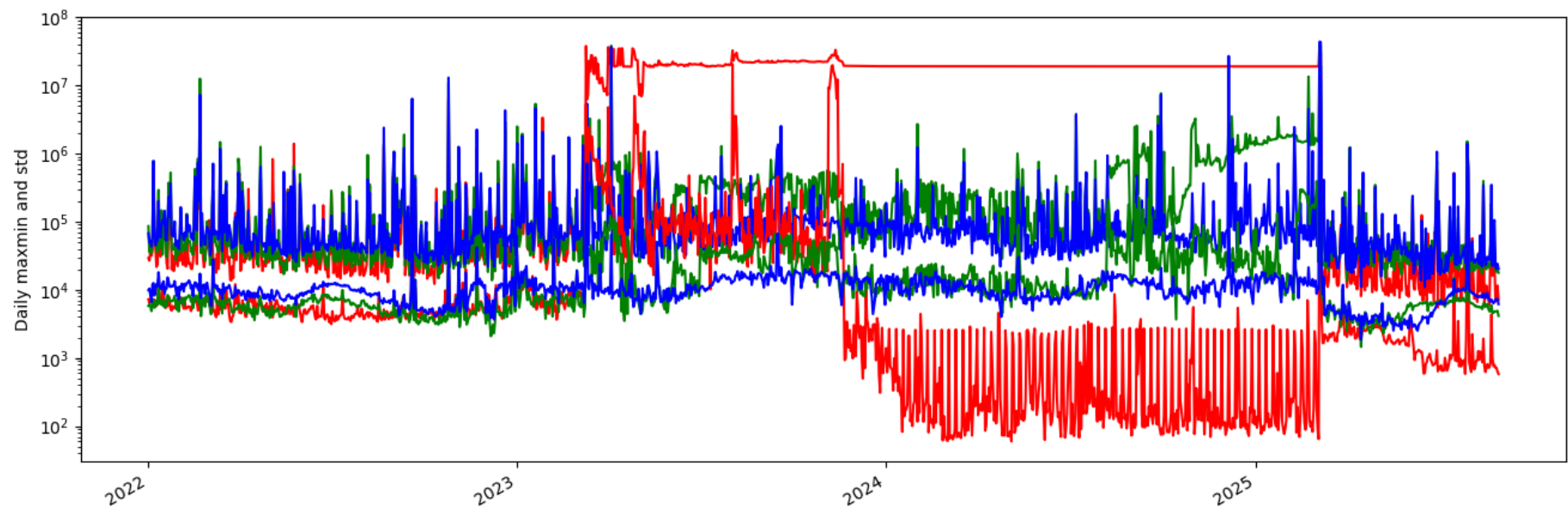
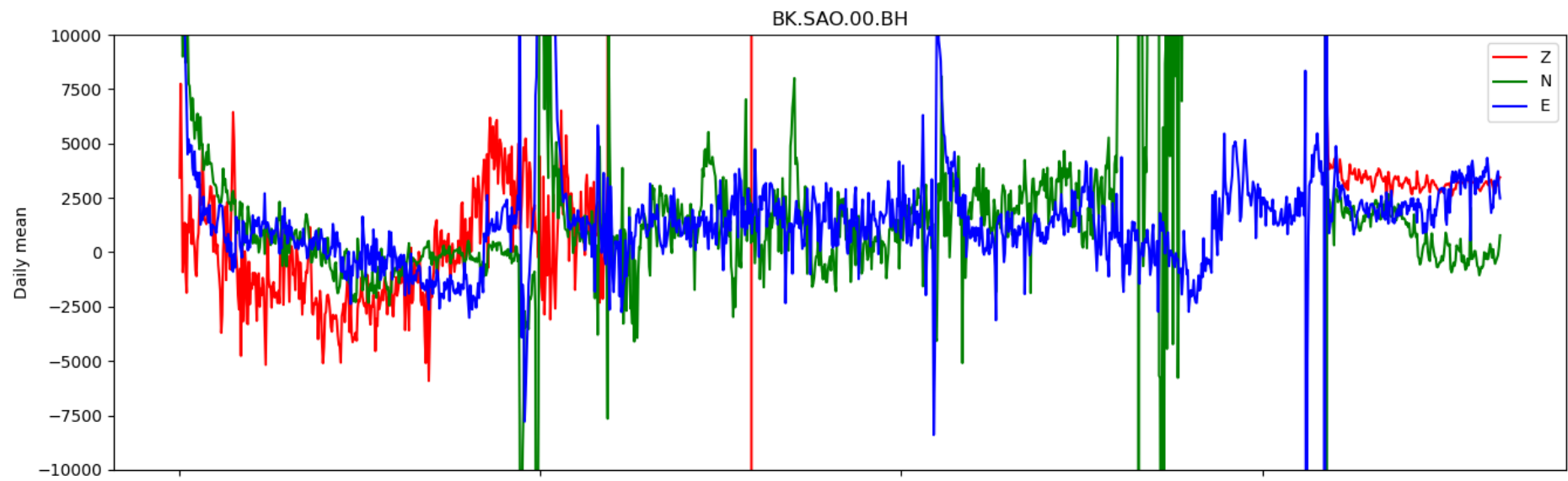


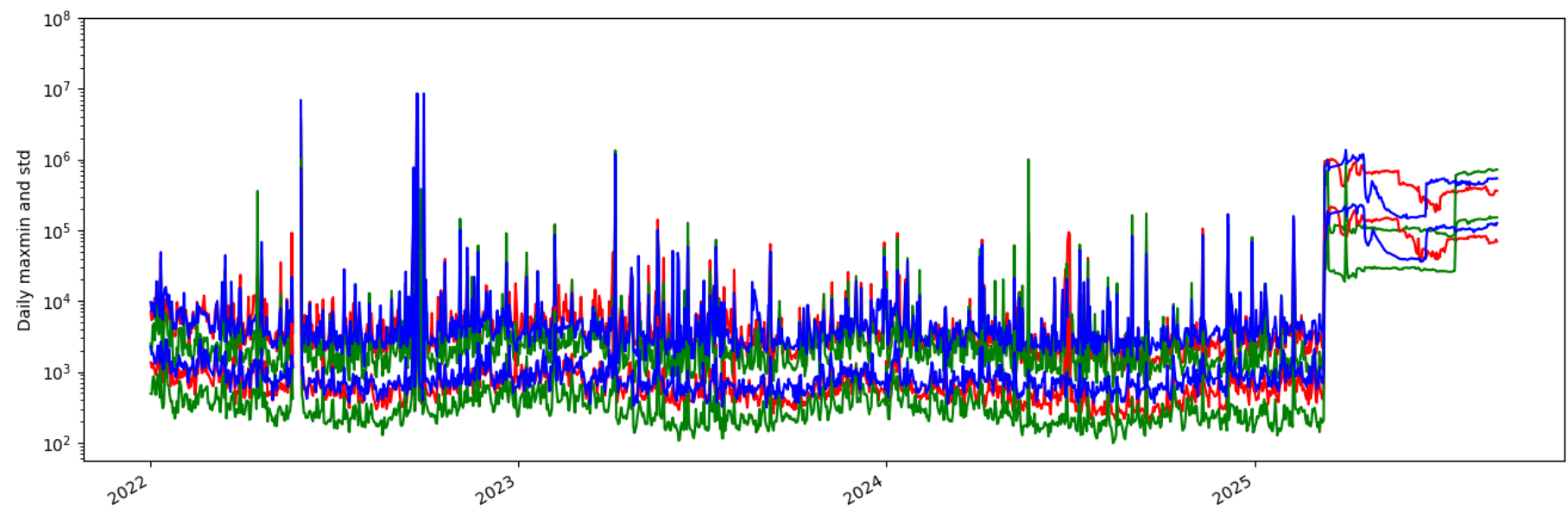
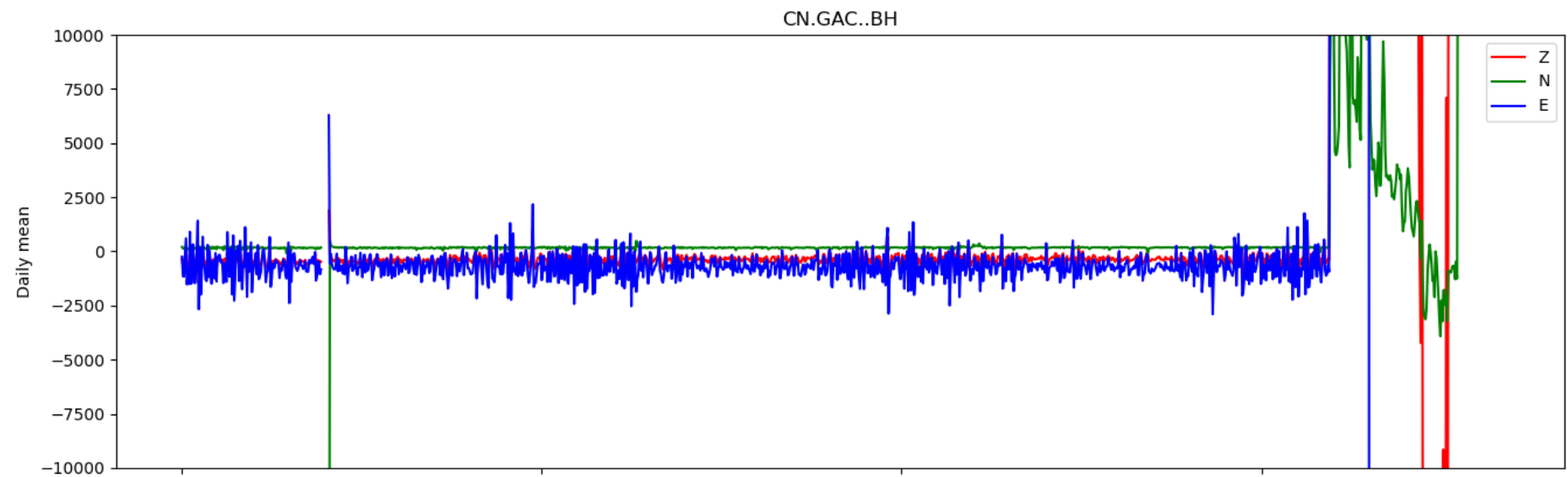
MEDIANS

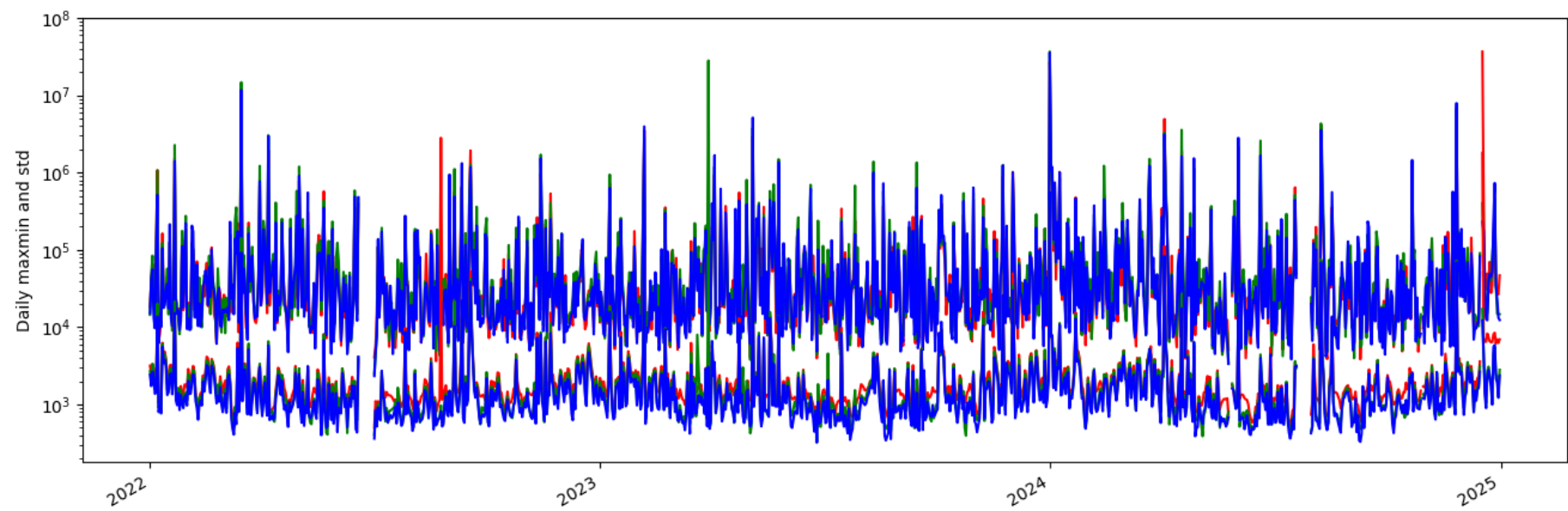
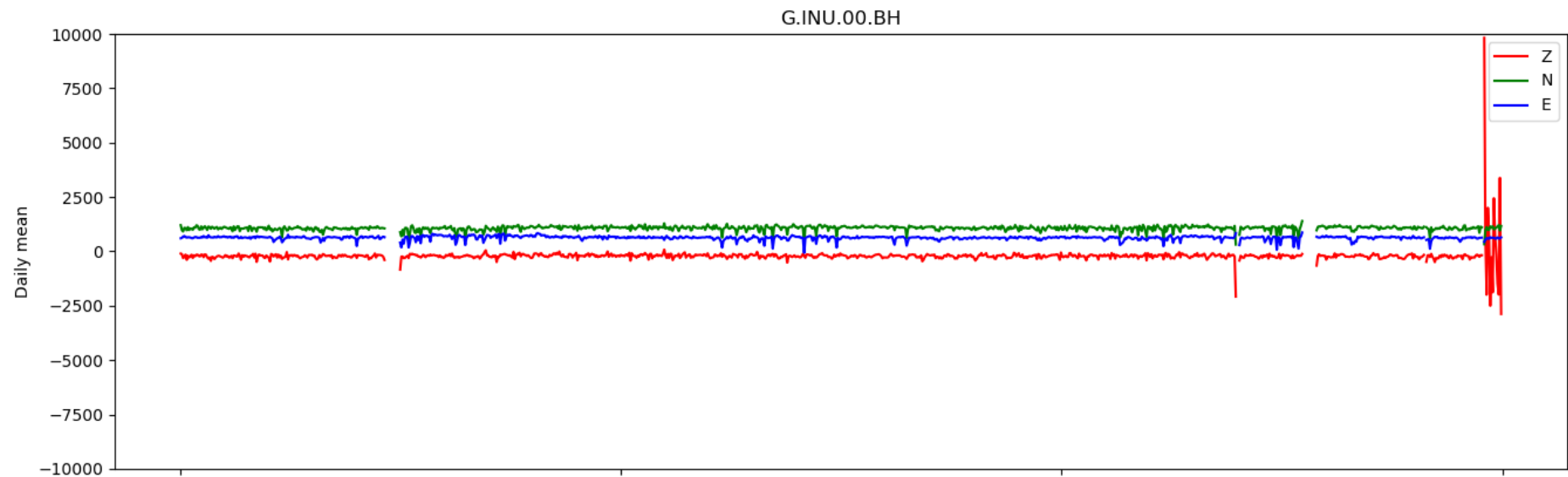


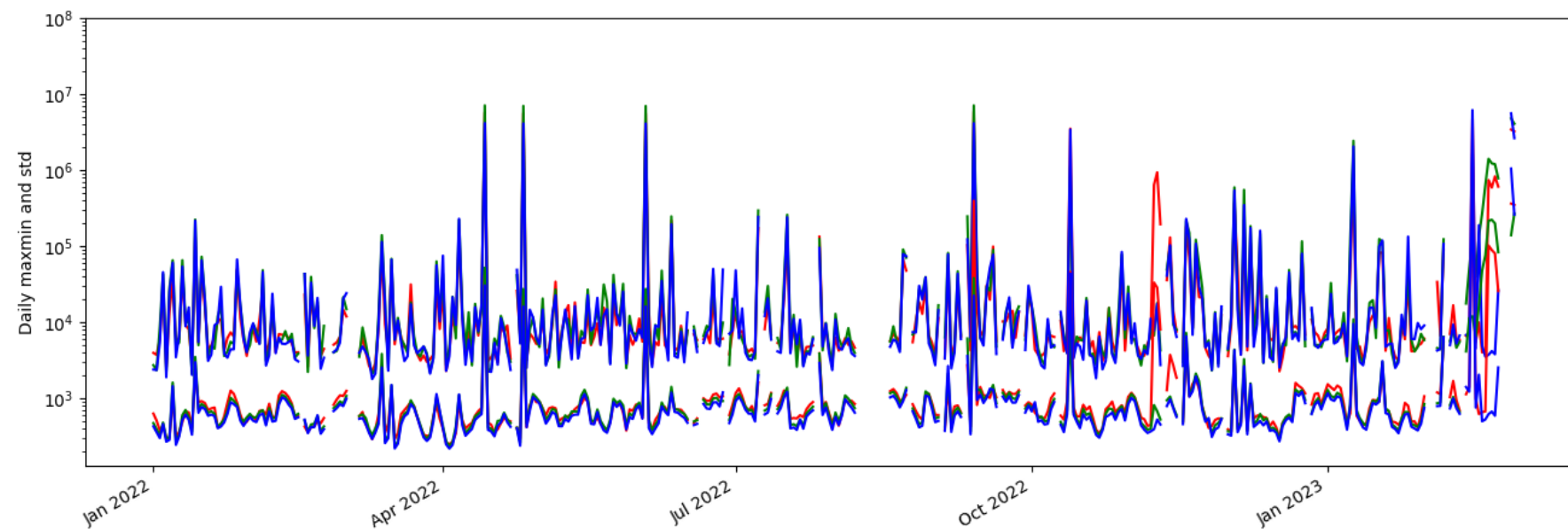
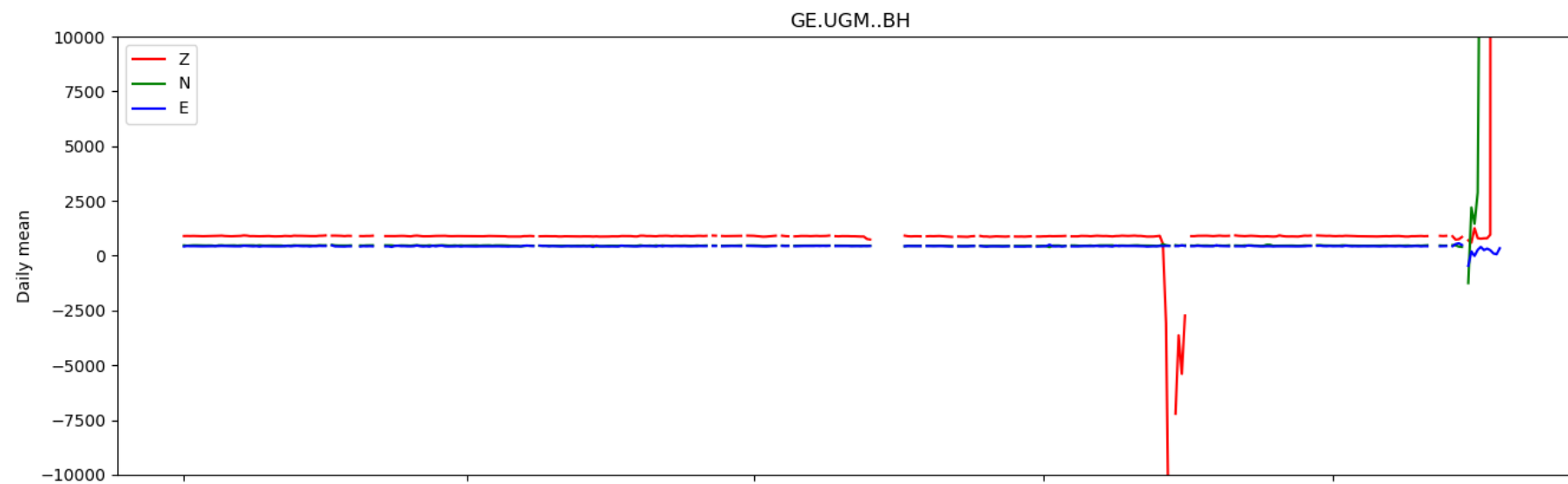
MAXMINs = MAX - MIN

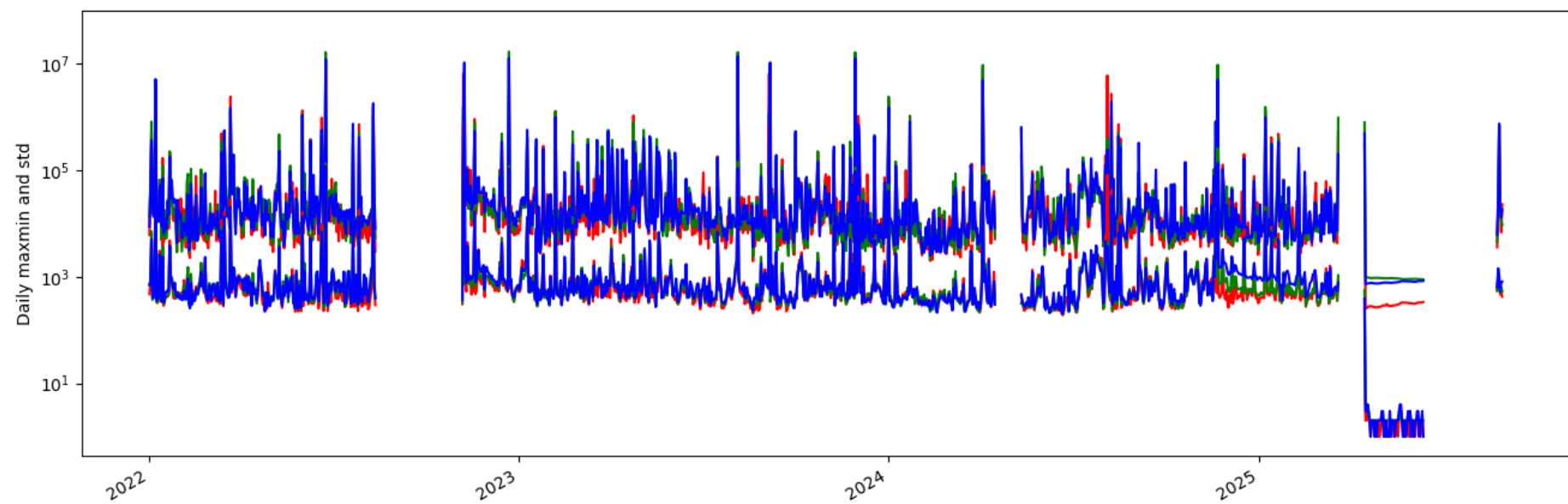
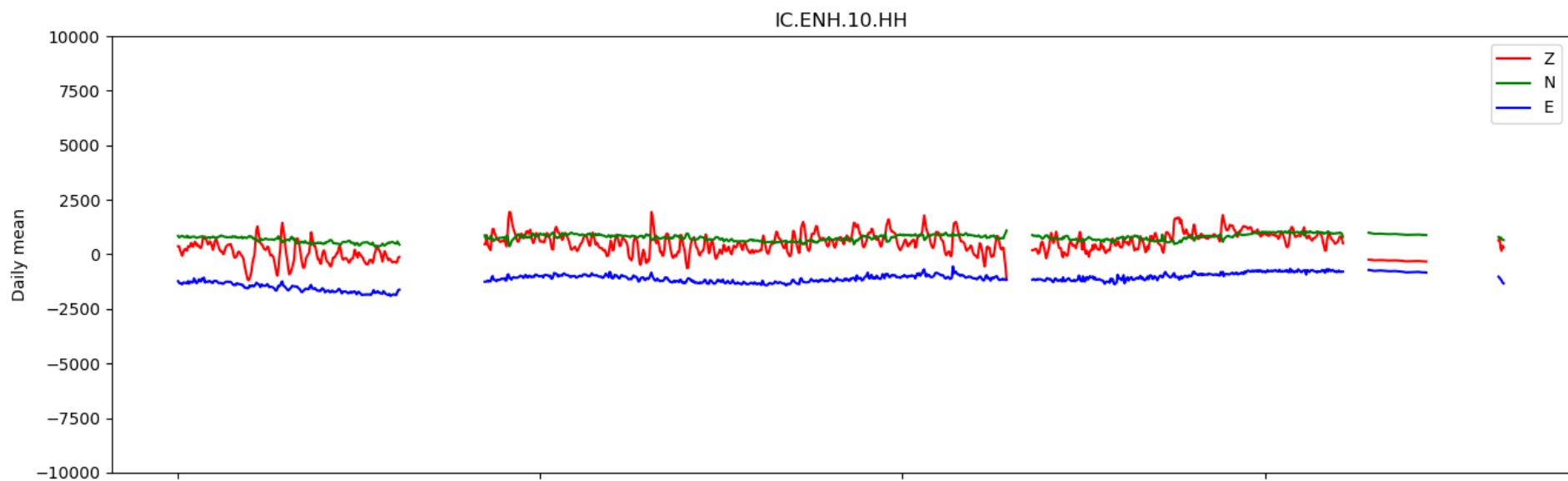
STDs











IDEAS

- preparing SoH report periodically (each year?) (and contacting operators?)
- quality metrics:
 - median (maxmin, std)
 - PPSD
 - data availability
- best way how to use EIDA & ErthScope QC tools
- other checks: sensor orientation, ...